

[6] Touch Sensor

DESIGN NOTE:

Table 3-2. Predefined Auto-off Delay (Active High Output)

Vt	Auto-off Delay (t _o)
Vss	Infinity (remain on until toggled to off)
Vdd	15 minutes
OUT	60 minutes

Table 3-3. Predefined Auto-off Delay (Active Low Output)

Vt	Auto-off Delay (t _o)
Vss	15 minutes
Vdd	Infinity (remain on until toggled to off)
OUT	60 minutes

DESIGN NOTE:

Output Polarity Selection

The output (OUT pin) of the QT1012 can be configured to have an active high or active low output by means of the output configuration resistor R_{op}. The resistor is connected between the output and either Vss or Vdd (see Figure 3-4 and Table 3-1). A typical value for R_{op} is 100 kΩ.

Table 3-1. Output Configuration

Name (V _{op})	Function (Output Polarity)
Vss	Active high
Vdd	Active low

DESIGN NOTE:
Cd2
decoupling
capacitor

DESIGN NOTE:
sense circuit
Cs [2nF - 60nF]
Rs [4k7 - 20k]

DESIGN NOTE:

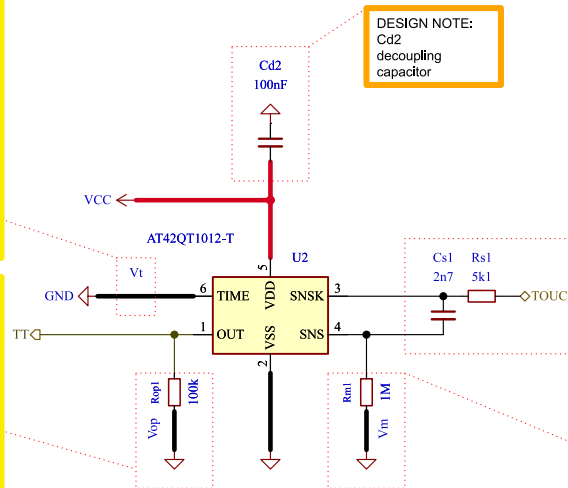
A single 1 MΩ resistor (R_m) is connected between the SNS pin and the logic level V_m to provide three auto-off functions: delay multiplication, delay override and delay retriggering. On power-up the logic level at V_m is assessed and the delay multiplication factor is set to x1 or x24 accordingly (see Figure 3-6 on page 12, Table 3-2 on page 12 and Table 3-3 on page 12). At the end of each acquisition cycle the logic level of V_m is monitored to see if an Auto-off delay override is required (see Section 3.11.5 on page 17). Setting the delay multiplier to x24 will decrease the key sensitivity. To compensate, it may be necessary to increase the value of C_s.

Table 3-4. Auto-off Delay Multiplier

V _m	Auto-off Delay Multiplier
Vss	t _o × 1
Vdd	t _o × 24

ACTIVE HIGH OUTPUT

AUTO DELAY OFF



Comments:

Company:

BaJRan imaginary industries

Variant:

CHECKED

Git Hash:

5ba43f3

Board Name:

3ART12 TOUCH

Project Name:

Sienkiewicz

Sheet Title:

Touch Sensor

File Name:

Section C - Touch sensor.kicad_sch

Designer:

Pawel Baran

Date:

2026-04-14

Revision:

2.1.1+ (Unreleased)

Sheet Path:

/Project Architecture/Section C - Touch sensor/

Reviewer:

Size:

A4

Sheet:

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